pR 4: Second year Computer Engineering class, set A of students like Vanilla Ice-cream and set B of students like butterscotch ice-cream. Write a program to store two sets using a linked list. compute and displaya) Set of students who like both vanilla and butterscotch b) Set of students who like either vanilla or butterscotch or not both c) Number of students who like neither vanilla nor butterscotch

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// Online C++ compiler (editor)
// Write and run C++ online using this editor.
#include <iostream>
#include <set>
using namespace std;

struct Node
{
    int rollNo;
    Node* next;
};

Node* createSet(int n)
{
    Node* head = NULL;
    Node* temp = NULL;

    for (int i = 0; i < n; i++)
    {
        int roll;
        cout << "Enter roll number: ";
        cin >> roll;

        Node* newNode = new Node;
        newNode->rollNo = roll;
        newNode->next = NULL;

        if (head == NULL)
            head = newNode;
        else
            temp->next = newNode;

        temp = newNode;
    }
    return head;
}
```

```
void displaySet(set<int> s)
{
    if (s.empty())
    {
        cout << "Empty set";
    }
    else
    {
        for (int x : s)
            cout << x << " ";
    }
    cout << endl;
}

int main()
{
    int n, m;
    set<int> A, B, both, either, allStudents;

    cout << "Enter number of students who like Vanilla: ";
    cin >> n;

    cout << "Enter roll numbers for Vanilla set:\n";
    Node* headA = createSet(n);

    cout << "Enter number of students who like Butterscotch: ";
    cin >> m;

    cout << "Enter roll numbers for Butterscotch set:\n";
    Node* headB = createSet(m);

    // Store linked list elements in sets
    for (Node* p = headA; p != NULL; p = p->next)
        A.insert(p->rollNo);

    for (Node* p = headB; p != NULL; p = p->next)
        B.insert(p->rollNo);

    // Find students who like both flavours
    for (int x : A)
        if (B.count(x))
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            both.insert(x);

    // Find students who like either flavour, but not both
    for (int x : A)
        if (!B.count(x))
            either.insert(x);

    for (int x : B)
        if (!A.count(x))
            either.insert(x);

    // Find all students who like at least one flavour
    allStudents = A;
    allStudents.insert(B.begin(), B.end());

    cout << "\nStudents who like both flavours: ";
    displaySet(both);

    cout << "Students who like either flavour, but not both: ";
    displaySet(either);

    // Assume total class strength is entered by the user
    int total;
    cout << "Enter total number of students in the class: ";
    cin >> total;

    int neither = total - (int)allStudents.size();

    cout << "Number of students who like neither flavour: "
         << neither << endl;

    return 0;
}
```